

Lab Tasks

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Lab #02

Activity- 1:

Write a Python script that uses a dictionary to store information about a person you know. Store their first name, last name, age, and the city in which they live. You should have keys such as `first_name`, `last_name`, `age`, and `city`. Print each piece of information stored in your dictionary.

```
person = {
    'first_name': 'Ali',
    'last_name': 'Khan',
    'age': 25,
    'city': 'Karachi'
}
print("Activity 1:")
for key, value in person.items():
    print(f"{key.capitalize()}: {value}")
print()
```

```
Activity 1:
First_name: Ali
Last_name: Khan
Age: 25
City: Karachi
```

Activity- 2:

Write a function that accepts a dictionary as an argument. If the dictionary contains replicate values, return an empty dictionary, otherwise, return a new dictionary whose values are now the keys and whose keys are the values.

```
def swap_dict(d):
    if len(set(d.values())) != len(d.values()):
        return {}
    return {v: k for k, v in d.items()}

print("Activity 2:")
sample_dict = {'a': 1, 'b': 2, 'c': 3}
print(swap_dict(sample_dict))
sample_dict_with_duplicates = {'a': 1, 'b': 2, 'c': 1}
print(swap_dict(sample_dict_with_duplicates))
print()
```

```
Activity 2:
{1: 'a', 2: 'b', 3: 'c'}
{}
```

Activity- 3:

A buffet-style restaurant offers only five basic foods. Think of five simple foods, and store them in a tuple.

- Use a for loop to print each food the restaurant offers.
- Try to modify one of the items, and make sure that Python rejects the change.

```
print("Activity 3:")
foods = ('Pizza', 'Burger', 'Pasta', 'Fries', 'Salad')
for food in foods:
    print(food)
foods[0] = 'Sushi'
print()
```

Activity 3:
 Pizza
 Burger
 Pasta
 Fries
 Salad

```
-----
TypeError                                Traceback (most recent call last)
Cell In[10], line 5
      3 for food in foods:
      4     print(food)
----> 5 foods[0] = 'Sushi'
      6 print()

TypeError: 'tuple' object does not support item assignment
```

Activity- 4:

If you could invite anyone to dinner. Make a list that includes at least three people you'd like to invite to dinner. Then use your list to print a message to each person, inviting them to dinner.

```
print("Activity 4:")
guests = ['Hamza', 'Sara', 'Ahmed']
for guest in guests:
    print(f"Dear {guest}, you're invited to dinner at my place!")
print()
```

Activity 4:
 Dear Hamza, you're invited to dinner at my place!
 Dear Sara, you're invited to dinner at my place!
 Dear Ahmed, you're invited to dinner at my place!

Activity- 5:

Changing Guest List: You just heard that one of your guests can't make the dinner, so you need to send out a new set of invitations. You'll have to think of someone else to invite.

- Modify your list, replacing the name of the guest who can't make it with the name of the new person you are inviting.
- Print a second set of invitation messages, one for each person who is still in your list.

```
print(f"{guests[1]} can't make it to dinner.")
guests[1] = 'Bilal'
for guest in guests:
    print(f"Dear {guest}, you're still invited to dinner at my place!")
print()
```

Activity 5:
 Fatima can't make it to dinner.
 Dear Hamza, you're still invited to dinner at my place!
 Dear Bilal, you're still invited to dinner at my place!
 Dear Ahmed, you're still invited to dinner at my place!

Activity- 6:

Write a program to read 6 numbers and create a dictionary having keys EVEN and ODD.

Dictionary's value should be stored in list. Your dictionary should be like: {'EVEN':[8,10,64], 'ODD':[1,5,9]}

```
print("Activity 6:")
numbers = []
print("Enter 6 numbers:")
for _ in range(6):
    numbers.append(int(input()))
result = {'EVEN': [], 'ODD': []}
for n in numbers:
    if n % 2 == 0:
        result['EVEN'].append(n)
    else:
        result['ODD'].append(n)
print(result)
print()
```

Activity 6:

Enter 6 numbers:

1
2
6
7
3
4

{'EVEN': [2, 6, 4], 'ODD': [1, 7, 3]}

Activity- 7:

Write a definition of a method count_now(places) to find and display those place names, in which there are more than 5 characters.

```
def count_now(places):
    print("Activity 7:")
    for place in places:
        if len(place) > 5:
            print(place)

count_now(['London', 'Tokyo', 'Istanbul', 'Karachi', 'New York'])
print()
```

Activity 7:

London
Istanbul
Karachi
New York

Activity- 8:

Write the following 2 functions: def

 ComputeOddSum(num): def

 ComputeEvenSum(num):

- The function ComputeOddSum find the sum of all odd numbers less than num.
- The function ComputeEvenSum find the sum of all even numbers less than num.

```
def ComputeOddSum(num):  
    return sum(i for i in range(num) if i % 2 != 0)  
  
def ComputeEvenSum(num):  
    return sum(i for i in range(num) if i % 2 == 0)  
  
print("Activity 8:")  
print("Odd sum under 10:", ComputeOddSum(10))  
print("Even sum under 10:", ComputeEvenSum(10))  
print()
```

Activity 8:
Odd sum under 10: 25
Even sum under 10: 20

Activity- 9:

Write a recursive function to get sum of all number from 1 up to give number. Example N = 5 Result must be sum (1+2+3+4+5) =15.

```
def recursive_sum(n):  
    if n <= 1:  
        return n  
    return n + recursive_sum(n - 1)  
  
print("Activity 9:")  
print("Sum up to 5:", recursive_sum(5))
```

Activity 9:
Sum up to 5: 15